

Megan K. Ayers

New Haven, CT | m.ayers@yale.edu | 208-914-4941

RESEARCH INTERESTS

Causal inference, environmental impact evaluation, experimental design, remote sensing, machine learning, text as data, computational social science, social and environmental applications of statistics

EDUCATION

Ph.D. in Statistics & Data Science, Yale University, Fall 2025 (expected)

- Advanced to candidacy May 2023
- Dissertation: *Causal Inference in Complex Settings: Innovations and Perspectives for Textual and Environmental Contexts*

M.A. in Statistics & Data Science, Yale University, 2024

B.A. in Mathematics, B.A. in Physics, Lewis & Clark College, 2019

- Summa Cum Laude, Phi Beta Kappa, Pi Mu Epsilon
- GPA: 3.98/4.00

PUBLICATIONS

PEER REVIEWED JOURNAL PUBLICATIONS

Ayers, M., Marlon, J. R., Ballew, M. T., Maibach, E. W., Rosenthal, S. A., Roser-Renouf, C., & Leiserowitz, A. (2024). Changes in Global Warming's Six Americas: An analysis of repeat respondents. *Climatic Change*, 177(6), 96. <https://doi.org/10.1007/s10584-024-03754-x>

Liu, T., Gezari, S., Ayers, M., Burgett, W., Chambers, K., Hodapp, K., Huber, M. E., Kudritzki, R.-P., Metcalfe, N., Tonry, J., Wainscoat, R., & Waters, C. (2019). Supermassive Black Hole Binary Candidates from the Pan-STARRS1 Medium Deep Survey. *The Astrophysical Journal*, 884(1), 36. <https://doi.org/10.3847/1538-4357/ab40cb>

Foord, A., Gültekin, K., Reynolds, M., Ayers, M., Liu, T., Gezari, S., & Runnoe, J. (2017). A Multi-wavelength Analysis of Binary-AGN Candidate PSO J334.2028+01.4075. *The Astrophysical Journal*, 851(2), 106. <https://doi.org/10.3847/1538-4357/aa9a39>

CONFERENCE & WORKSHOP PUBLICATIONS

Ayers, M.^{*}, Sanford, L.^{*}, Roberts, M. E., & Yang, E. (2024). Discovering influential text using convolutional neural networks. *Findings of the Association for Computational Linguistics*, pages 12002–12027, Bangkok, Thailand and virtual meeting.

^{*}Equal contribution

Gordon, M., Ayers, M., Stone, E., & Sanford, L. (2023). Remote Control: Debiasing Remote Sensing Predictions for Causal Inference. *Proc. Int. Conf. Learn. Represent. Workshops*, Kigali, Rwanda and virtual meeting.

PREPRINTS

Ayers, M.^{*}, Sanford, L.^{*}, Gardner, W., & Kuebbing, S. (2025). Causal Carbon: Baselines and Additivity with Potential Outcomes. https://doi.org/10.31219/osf.io/5pcuh_v2

^{*}Equal contribution

Sanford, L.^{*}, Ayers, M.^{*}, Gordon, M.^{*}, & Stone, E. (2025). Adversarial Debiasing for Unbiased Parameter Recovery. <https://www.arxiv.org/abs/2502.12323>

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TEACHING & MENTORING EXPERIENCE

Teaching Fellow, Yale University, 2021-2024

- Statistical Case Studies (Fall 2022, Fall 2024)
- Data Analysis (Spring 2022, Spring 2023)
- Statistical Computing (Fall 2022)
- Linear Models (Fall 2021)

Grader without Contact, Yale University, 2023-present

- Statistical Consulting (Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025)

College Teaching Preparation Program Workshops, Yale University

- Advanced Teaching two-part workshop series on “How We Learn”, Spring 2025
- Advanced Teaching Workshop on Course Design, Fall 2024
- Intermediate Teaching Workshop on Teaching First-Generation and Low-Income Students, Fall 2024
- Intermediate Teaching Workshop on Grading and Rubrics, Fall 2024
- Intermediate Teaching Workshop on Mental Health in the Classroom, Fall 2024
- Intermediate Teaching Workshop on Teaching Quantitative Reasoning, Spring 2024
- Fundamentals of Equitable Teaching four-part workshop series, Spring 2024

Peer Tutor in Math & Physics, Lewis & Clark College, 2016-2019

Math Department Grader, Lewis & Clark College, 2016-2019

- Partial Differential Equations with Applications (Spring 2019)
- Linear Algebra (Fall 2018)
- Differential Equations (Fall 2017)
- Calculus I (Fall 2016, Spring 2017)

FELLOWSHIPS & AWARDS

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| 2023-2024 | Planetary Solutions Project seed grant from the Climate Impact Innovation Fund and the Gordon Data and Environmental Sciences Research Grant (research assistant funding recipient), Yale University |
| 2023 | Graduate Student Assembly Conference Travel Fellowship, Yale University |
| 2020 | Sterling Prize Fellowship, Yale University |
| 2018 | Shannon T. O’Leary Award, Lewis & Clark College |

PRESENTATIONS

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| 2024 | “Discovering influential text using convolutional neural networks,” Association for Computational Linguistics , August 12-14 and August 22, Bangkok, Thailand and virtual meeting. (Virtual poster) |
| | “A New Perspective for Forest Carbon Offsetting through the Potential Outcomes Framework,” American Causal Inference Conference Annual Meeting , May 14-17, Seattle, WA. (Poster) |
| | “Site Matching Baselines with Real and Synthetic Controls,” Workshop on Increasing Consensus on Best Practices for Additionality & Baselines in Forest Carbon Crediting Accounting , April 11-12, Yale University. (Oral) |
| 2023 | “Assessing Leakage from Carbon Offset Projects,” Google Geo for Good Summit , October 10-12, Mountain View, CA. (Oral) |

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“Remote Control: Debiasing Remote Sensing Predictions for Causal Inference,”
The Workshop in Environmental Economics and Data Science, September 15-16,
Portland, OR. (Oral, presented by a colleague for health reasons)

“Remote Control: Debiasing Remote Sensing Predictions for Causal Inference,” **ICLR
Workshop on Machine Learning for Remote Sensing**, May 5, Kigali, Rwanda and
Virtual meeting. (Virtual oral)

“Remote Control: Debiasing Remote Sensing Predictions for Causal Inference,” **ICLR
Workshop on Tackling Climate Change with Machine Learning**, May 4, Kigali,
Rwanda and virtual meeting. (Virtual poster)

RESEARCH & PROFESSIONAL EXPERIENCE

Yale Center for Environmental Law & Policy

New Haven, CT

Summer Research Assistant

June-August 2022

- Designed the “EPI Weighting Explorer” Tableau dashboard to accompany the 2022 Environmental Performance Index Report, allowing readers to explore EPI scores and rankings when indicator weights are adjusted. As of August 2024, the dashboard has reached over 11,000 views

Yale Law School Legal Services Organization

New Haven, CT

Statistical/Data Analyst

November 2020-March 2021

- Collaborated with law students in the Veterans Legal Services Clinic to prepare a report examining ethnic/racial disparities in congressional nominations to the U.S. Military Service Academies

Cascade Data Labs

Portland, OR

Data Analyst

June 2019-May 2020

- Utilized eCommerce data to provide business insights to clients through interactive data tools and ad hoc analyses
- Communicated technical concepts to clients with varying backgrounds and collaborated with teams of stakeholders to tailor data products to clients’ needs

Yale University Sackler NSF-REU

New Haven, CT

Research Intern (Emonet Lab)

May-August 2018

- Developed MATLAB code to conduct and analyze agent-based simulations of E. coli cells

Maria Mitchell Observatory NSF-REU

Nantucket, MA

Research Intern

May-August 2017

- Constructed and analyzed quasar energy distributions for research on supermassive black hole binaries
- Operated 24” research telescope to contribute 14 nights of data for observational research of variable star MWC 349
- Engaged with public during open nights attended by over 2250 guests and led weekly informational tours of historical observatory

SERVICE

Ad-hoc reviewer, *Climatic Change*, 2024

Workshop organizer, Workshop on Increasing Consensus on Best Practices for Additionality & Baselines in Forest Carbon Crediting Accounting, 2024

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Undergraduate Learning Assistant hiring coordinator, Yale University Department of Statistics & Data Science, 2021-2022